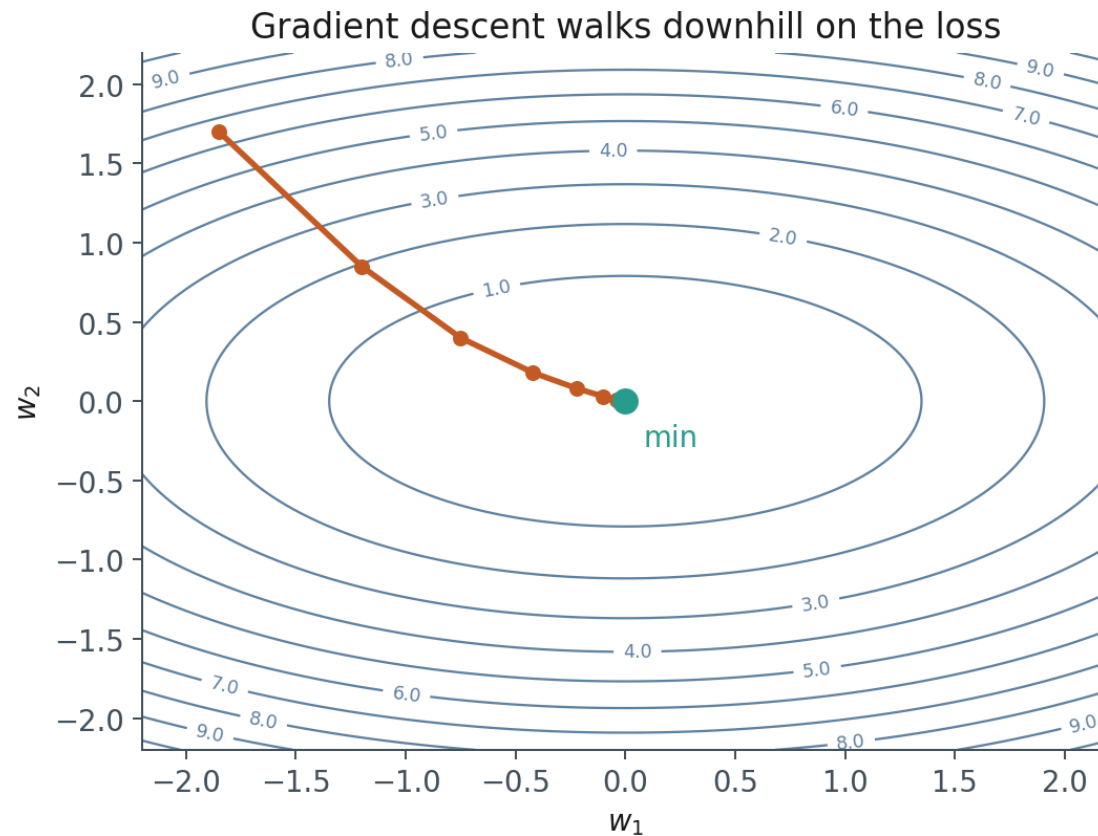


Training is walking downhill

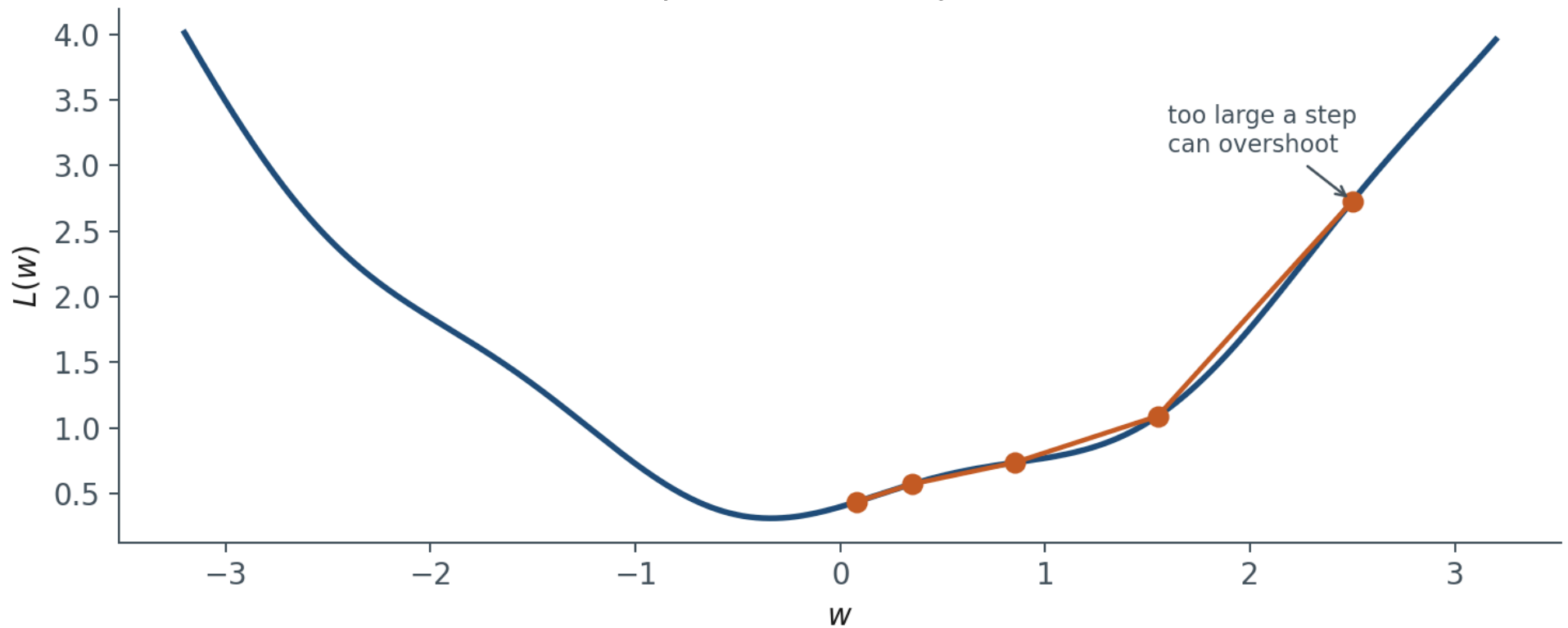
- Pack weights into θ . Loss $L(\theta)$ is a scalar.
- The gradient points uphill. We step the other way.



The update and the learning rate

- $\theta \leftarrow \theta - \eta \nabla L(\theta)$.
- Too small: crawl. Too large: overshoot.
- SGD estimates the gradient on a minibatch.

Update: $w \leftarrow w - \eta \nabla L(w)$



Forward, loss, backward, update

- Keep activations: the backward pass needs them.
- Backprop is the chain rule. PyTorch does it if the graph is differentiable.
- Pretraining and fine-tuning are this loop with different data and L .

Training a net is this loop. Backprop is just the chain rule.

